CLINICAL · EAR

Cholesteatoma

A cholesteatoma is skin in the wrong place: keratinising squamous epithelium growing inside the middle ear and mastoid, shedding keratin it cannot clear, expanding and eroding bone as it goes. Drops will not cure it. The operation aims for a safe, dry ear first and hearing second, and the follow-up lasts for years.

AT A GLANCE

- Keratinising skin in the middle ear that erodes bone.
- Scanty foul discharge with an attic or marginal defect.
- CT maps the disease; non-EPI MRI finds recurrence.
- Surgery is the treatment: safe ear first, hearing second.
- Assume a lateral canal fistula until shown otherwise.
- EAONO/JOS staging; endoscopic ear surgery and mastoid obliteration are growing; DWI misses disease under about 3 mm.

1 CHAPTER ONE Overview

Skin in the wrong place.

1.1 Definition

A cholesteatoma is a three-dimensional collection of keratinising stratified squamous epithelium in the middle ear cleft, with accumulated keratin debris in its centre. Despite the name it contains no cholesterol and is not a tumour; *keratoma* would be a better word.

1.2 Structure

1.2.1 Matrix

The living wall: keratinising squamous epithelium resting on a layer of inflamed connective tissue, the *perimatrix*.

1.2.2 Contents

White, flaky or pearly keratin debris, often infected and foul smelling.

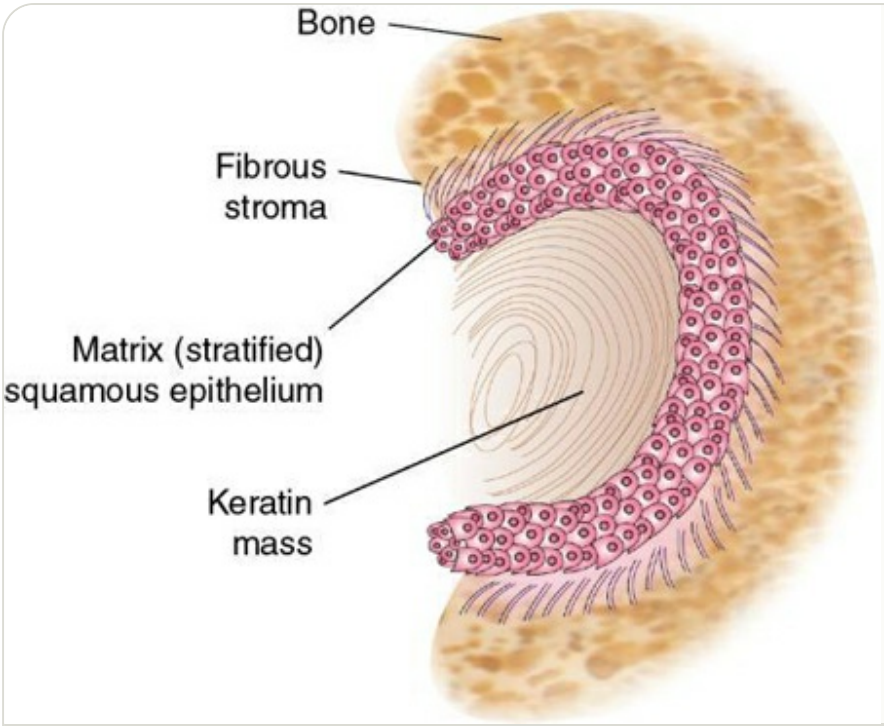


FIGURE 1 Structure of a cholesteatoma: a matrix of keratinising epithelium and perimatrix, enclosing keratin debris.

Dhingra, fig. 11.1

1.3 Why it matters

It is the *unsafe* chronic ear. Cholesteatoma destroys the ossicles, the labyrinth, the facial canal and the bony plates over the dura and sigmoid sinus, and it causes most of the serious complications of chronic otitis media.

2 CHAPTER TWO Classification and genesis

Where it comes from.

2.1 Congenital cholesteatoma

A white mass behind an intact drum, usually anterosuperior, in a child with no history of otitis media. It arises from epithelial rests left in the temporal bone, and may also occur in the petrous apex and the cerebellopontine angle.

2.2 Primary acquired cholesteatoma

The commonest type. A retraction pocket of the pars flaccida, drawn in by negative middle ear pressure, deepens into the attic; keratin accumulates in it, the neck narrows, and it becomes a self-perpetuating sac.

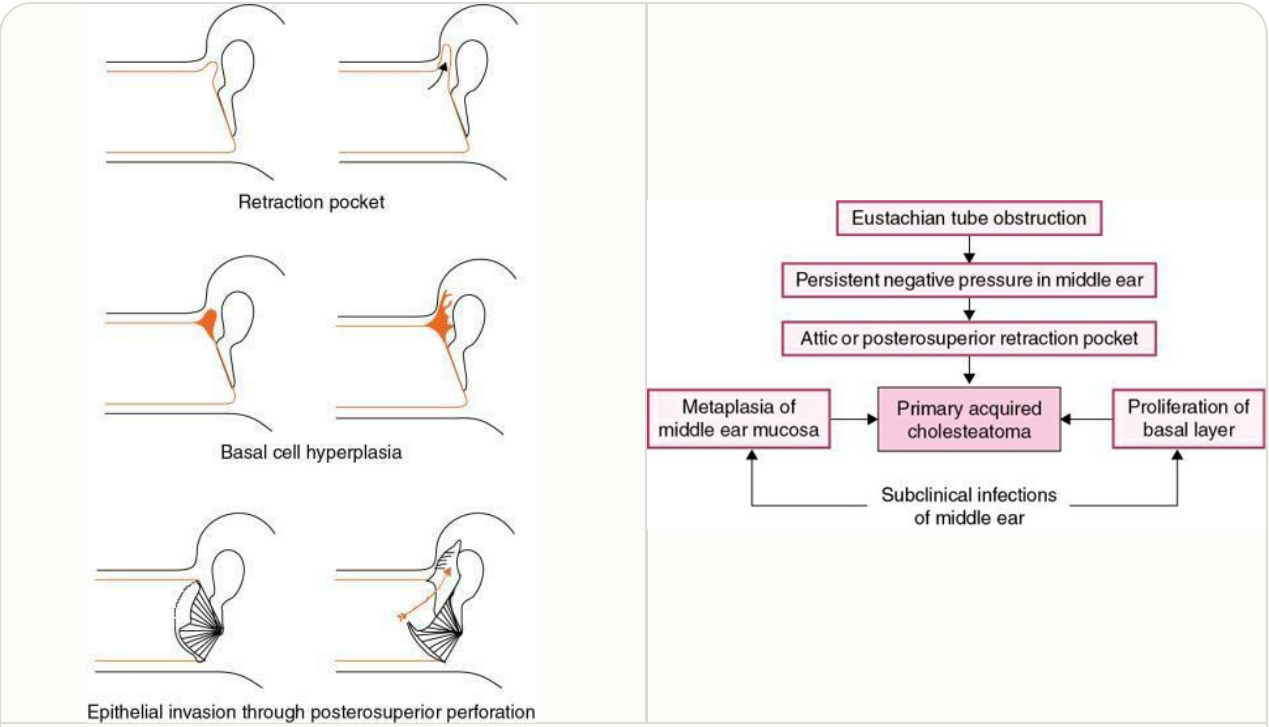


FIGURE 2 First, the theories of genesis: a retraction pocket, basal cell hyperplasia and epithelial invasion through a posterosuperior perforation. Second, the sequence leading to primary acquired cholesteatoma.

Dhingra, figs 11.2 and 11.3

2.3 Secondary acquired cholesteatoma

Squamous epithelium grows into the middle ear through a marginal or posterosuperior perforation, or is implanted by trauma or surgery.

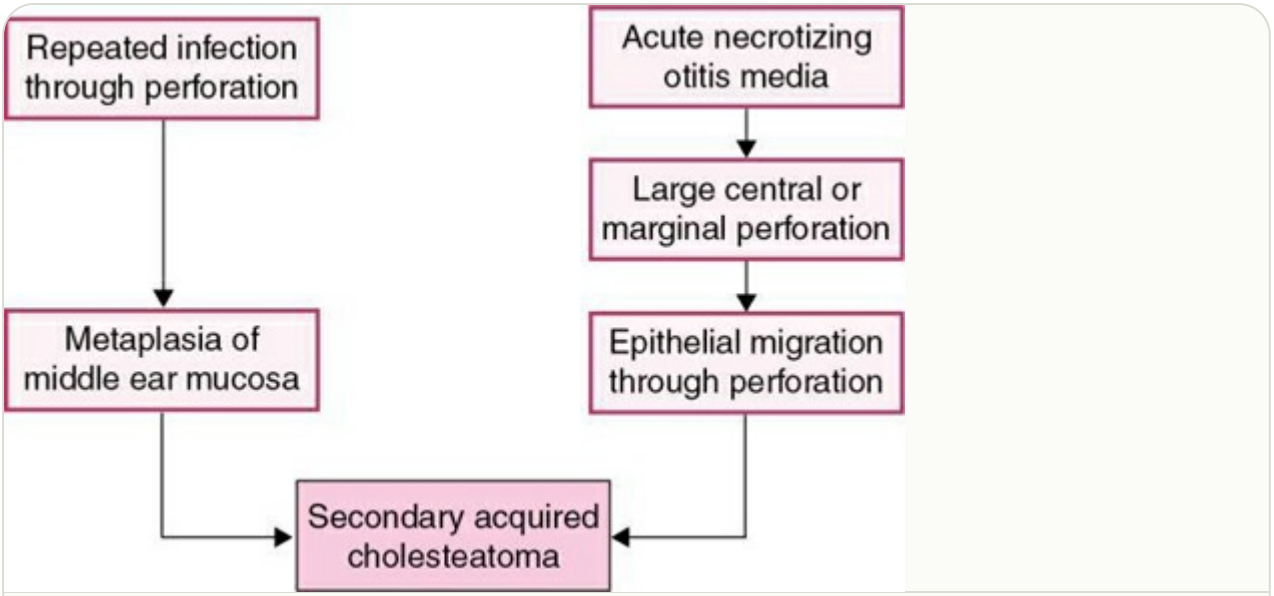


FIGURE 3 Pathways to secondary acquired cholesteatoma: metaplasia of the middle ear mucosa, and epithelial migration through a large central or marginal perforation.

Dhingra, fig. 11.4

2.4 Theories of genesis

- **Cell rests:** for congenital cholesteatoma.
- **Invagination or retraction (Wittmaack):** for primary acquired.
- **Basal cell hyperplasia (Ruedi):** prickle cells of the pars flaccida invade the subepithelial tissue.
- **Epithelial invasion or migration (Habermann):** through a marginal perforation.
- **Squamous metaplasia (Sade):** of the middle ear mucosa after chronic inflammation.

2.5 Retraction pockets

Retraction of the pars tensa is staged by how deep it goes.

- **Stage I:** retracted, not touching the incus.
- **Stage II:** touching the incus.
- **Stage III:** lying on the promontory (atelectasis), but can be lifted off with suction.
- **Stage IV:** adherent to the promontory (adhesive otitis), with no middle ear space.

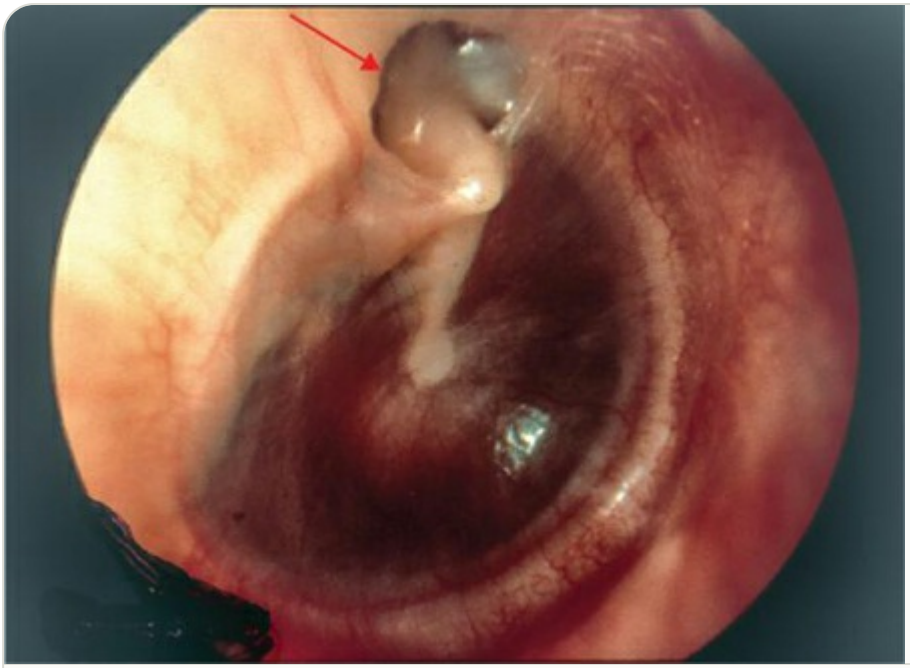


FIGURE 4 A retracted drum with an attic retraction pocket (arrow).
Dhingra, fig. 11.6

VENTILATION OF THE ATTIC

The attic is ventilated through the narrow anterior and posterior tympanic isthmus. When the isthmus is blocked by swollen mucosa, the attic and Prussak's space become isolated and the pars flaccida retracts, which is why attic cholesteatoma can coexist with a normally ventilated mesotympanum.

3 CHAPTER THREE Pathology

How it destroys bone.

3.1 Bone erosion

Collagenases and other enzymes from the matrix and perimatrix, osteoclast activation by inflammatory cytokines, and infection all contribute. The mastoid is usually sclerotic.

WHERE THE BOOKS DIFFER · HOW BONE IS ERODED

Dhingra	Lee

Attributes erosion to enzymes from osteoclasts and inflammatory cells, and rejects pressure necrosis.	Includes pressure along with the inflammatory mediators.

3.2 Routes of spread

The cholesteatoma follows the path of least resistance: from Prussak's space into the attic, then through the aditus to the antrum and mastoid; or posteriorly into the sinus tympani and facial recess, the sites hardest to clear at surgery.

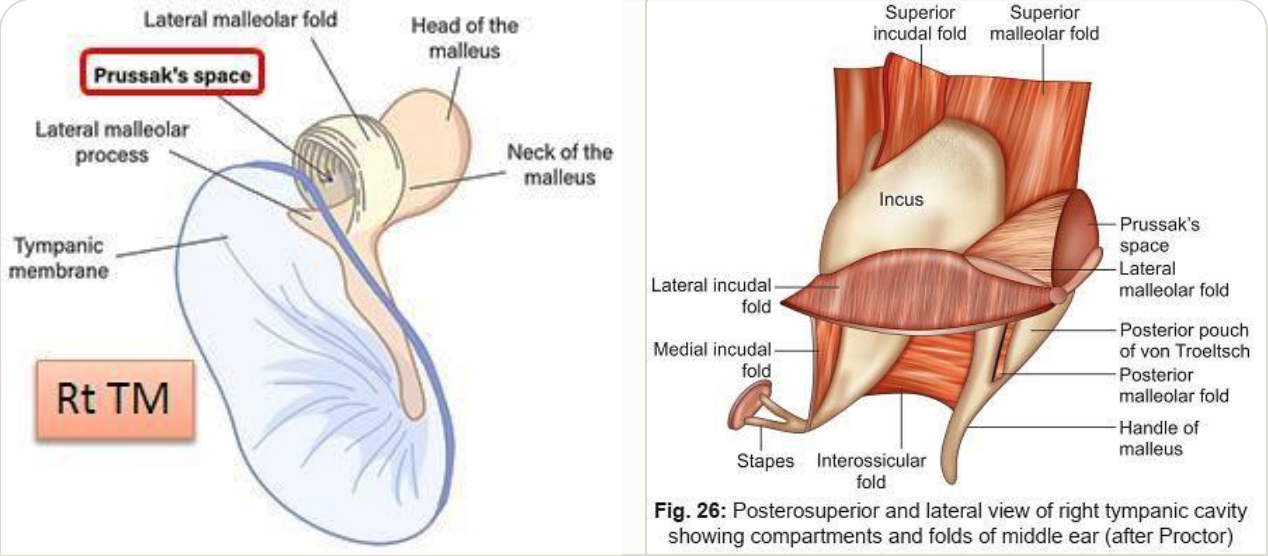


FIGURE 5 Prussak's space, where attic cholesteatoma begins, and the folds and compartments that guide its spread.

Anatomy atlas · ENT Steps p. 54

3.3 Hidden sites for residual disease

Sinus tympani, facial recess, the undersurface of the scutum, and the anterior epitympanic recess.

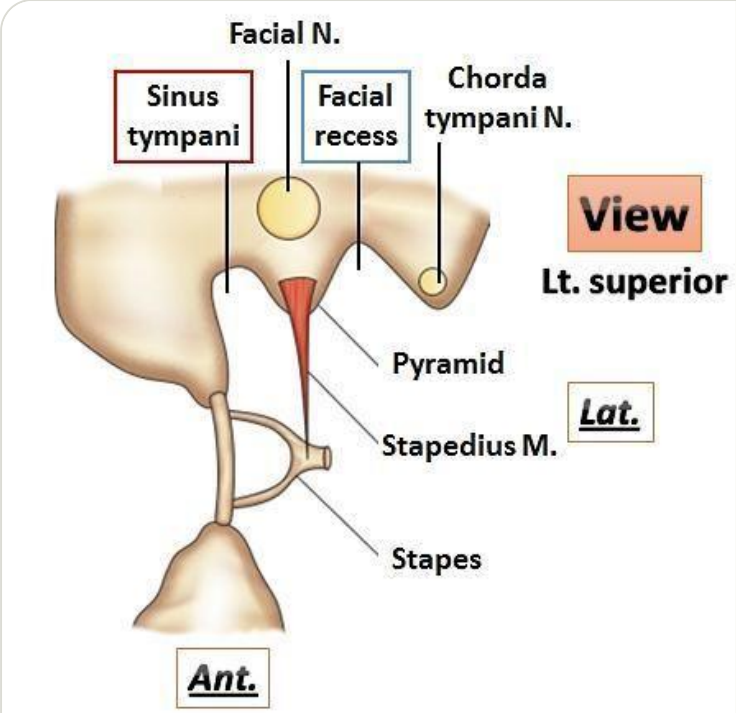


FIGURE 6 The posterior wall of the middle ear: the facial recess and sinus tympani, the commonest hiding places for residual disease.

Anatomy atlas · ENT Steps p. 46

4 CHAPTER FOUR Clinical features

The scanty, foul, unsafe ear.

4.1 Symptoms

4.1.1 Discharge

Scanty, foul smelling and sometimes bloodstained, from infected keratin and granulation.

A DISCHARGE THAT SUDDENLY STOPS

Pus may have been blocked and may be tracking inwards. Ask about headache, fever, vertigo and facial weakness.

4.1.2 Hearing loss

Conductive, or mixed when the cochlea is involved. A cholesteatoma that bridges an eroded chain can transmit sound well: the *cholesteatoma hearer*, who must be

warned that hearing may worsen after surgery.

4.1.3 Symptoms of complications

Earache, headache, vertigo, facial weakness, fever and vomiting, neck stiffness, double vision or unsteadiness, or a listless child who will not feed.

4.2 Signs

- An attic perforation or posterosuperior marginal perforation.
- White keratin flakes or a pearly mass.
- Erosion of the scutum or posterosuperior canal wall.
- A red, fleshy polyp that bleeds when touched.
- A positive fistula test in some.

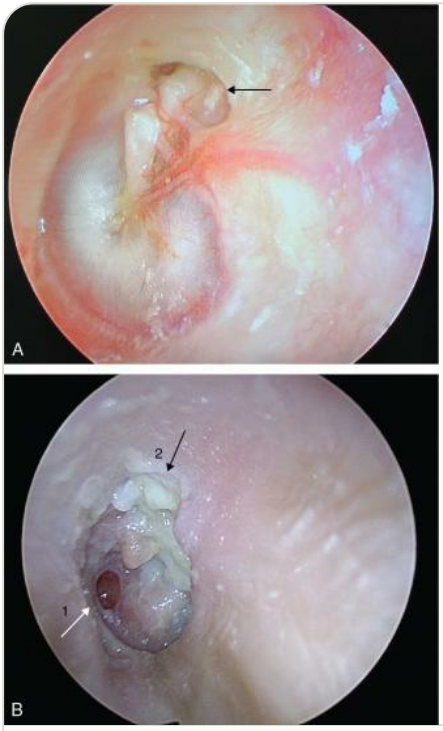


FIGURE 7 (A) An attic retraction pocket. (B) A double perforation, in the pars tensa and in the attic.
Dhingra, fig. 11.13

A BLUE DRUM

A blue drum with dark fluid behind it suggests cholesterol granuloma, not cholesteatoma.

5 CHAPTER FIVE Investigations

Map the disease before you operate.

5.1 Examination under the microscope

Clean with suction; look for keratin, the extent of the defect and polyps.

5.2 Fistula test

Pressing on the tragus or using a Siegle speculum causes vertigo and nystagmus if the lateral canal is eroded. It is specific but insensitive, positive in only 30 to 50% of fistulae. Do it in every case.

5.3 Audiogram

Measures conductive and sensorineural components before surgery.

5.4 High-resolution CT

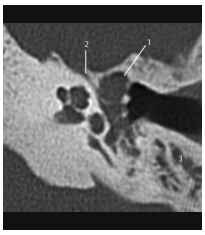
Shows the extent of disease and the state of the scutum, ossicles, tegmen, lateral semicircular canal, facial canal and sigmoid plate, and the anatomy for surgery.

5.5 Non-echo-planar diffusion-weighted MRI

Keratin restricts diffusion and shows bright. It detects residual or recurrent disease after canal wall up surgery, often replacing a second look, and helps when an intracranial complication is suspected.

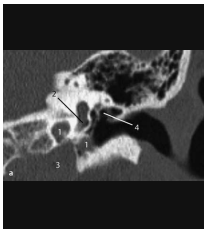
IMAGING IN THE APP 5 cases

Acquired and congenital cholesteatoma, a labyrinthine fistula, intracranial extension and canal cholesteatoma.



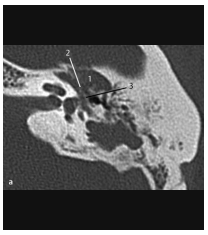
Cholesteatoma of the Middle Ear

Infection & inflammation · 12 images



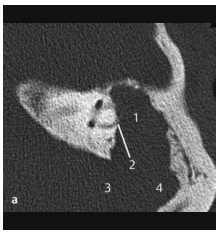
Congenital Cholesteatoma

Congenital · 11 images



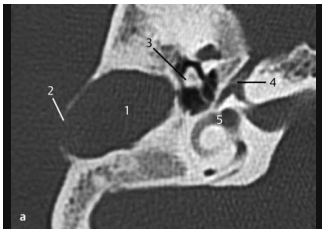
Labyrinthine Fistula due to Cholesteatoma

Infection & inflammation · 11 images



Cholesteatoma with Intracranial Involvement

Infection & inflammation · 7 images



Inclusion Cholesteatoma and Atresia of the External Auditory Canal

Congenital · 3 images

CHAPTER SIX

6 Management

Surgery is the treatment.

6.1 Aims

In order: eradicate disease and make the ear safe; obtain a dry ear; preserve or restore hearing.

6.2 Conservative care

Regular microscopic cleaning suits small accessible disease, patients over 65 or unfit for surgery, or those who refuse an operation.

6.3 Choice of operation

6.3.1 Canal wall up

The posterior canal wall is kept (combined approach tympanoplasty). The ear looks normal, swimming is allowed and hearing aids are easy to fit, but residual disease is commoner, so a second look at about six months or MRI follow-up is needed.

6.3.2 Canal wall down

The posterior canal wall is removed and the attic, antrum and mastoid are exteriorised into one cavity (modified radical or radical mastoidectomy). Exposure is better and recurrence lower, but the cavity needs lifelong cleaning, water precautions, and aids are harder to fit.

CANAL WALL UP OR DOWN

	CANAL WALL UP	CANAL WALL DOWN
Appearance	Normal meatus	Wide meatus and cavity
Residual disease	Higher	Lower
Second look	Often needed	Not routine
Cavity care	None	Regular cleaning
Swimming	Allowed	Limited
Hearing aid	Easy	Harder

Choose canal wall up only when the patient can return for long-term follow-up.

6.3.3 Endoscopic ear surgery

Transcanal endoscopes suit limited attic and mesotympanic disease, and see around corners into the sinus tympani and anterior attic without a postauricular incision. Disease reaching the antrum or mastoid needs a combined approach. The surgeon works one-handed, and bleeding and heat from the light source are drawbacks. Residual disease found by the endoscope after apparent microscopic clearance ranges from 17 to 38%.

OPERATIVE GUIDES IN THEATRE 3 operations

Mastoid Surgery
Mastoidectomy

Ossicular Chain Reconstruction
Ossicular reconstruction

Cholesterol Granulomas and Congenital Epidermoid Tumors of the Temporal Bone
Congenital and petrous disease

6.4 Lateral canal fistula

Present in 7 to 10% of cholesteatomas, 90% of them in the lateral canal; only about 60% are seen on preoperative CT. Assume one in every case.

- Small fistula: the matrix is often removed at the end of the operation and the fistula sealed.
- Large fistula or one in an only hearing ear: leave the matrix on it and convert to canal wall down.
- If the endosteum tears, cover it immediately.
- Never strip the matrix off a cochlear fistula.

6.5 Hearing reconstruction

Ossiculoplasty, at the first stage or the second look.

7 CHAPTER SEVEN Complications

Why it is called unsafe.

7.1 Intratemporal

- Ossicular erosion, usually the incus first.
- Labyrinthine fistula and labyrinthitis.
- Facial palsy, usually from the tympanic segment.
- Mastoiditis and abscesses.

7.2 Intracranial

Extradural and subdural abscess, meningitis, brain abscess, lateral sinus thrombosis and otitic hydrocephalus, through an eroded tegmen or sinus plate.

AT SURGERY

Facial canal dehiscence and tegmen dehiscence often accompany a fistula. The epineurium does not need opening unless disease has invaded the nerve.

CURRENT PRACTICE (OXFORD TEXTBOOK)

Staging. The EAONO/JOS system, from European and Japanese otological societies, divides the middle ear and mastoid into sites, with the *sinus tympani* and *protympanum* counted as difficult-access sites where residual disease is most often left. Stage I disease occupies one site, stage II two or more, stage III has an extracranial complication (facial palsy, labyrinthine fistula, postauricular abscess, canal wall destruction), and stage IV has intracranial spread. Using it allows outcomes to be compared between surgeons and techniques.

Endoscopes. A rigid 0° endoscope in clinic shows the full depth of a retraction pocket and documents it photographically. *Total endoscopic ear surgery* through the canal suits disease confined to the middle ear and attic: the wide angle shows the sinus tympani, facial recess and anterior attic better than the microscope, without a postauricular incision, but it is more demanding and its long-term recidivism figures are still emerging.

Canal wall up or down. Neither is a gold standard, and evidence on hearing outcomes is equivocal. Canal wall up surgery is often staged, with a planned second stage some months later (the Oxford Textbook suggests 3 to 6) to remove residual disease and rebuild the ossicular chain. About 40% of open cavities discharge or are intolerant of water, which is why *mastoid obliteration* (bone pâté, muscle flaps or bioactive glass, with cartilage canal wall reconstruction) is increasingly used: it avoids a cavity and lowers recidivism.

Follow-up imaging. *Non-echo-planar diffusion-weighted MRI* has a sensitivity of about 90% and specificity of about 95% for residual cholesteatoma, but disease smaller than about 3 mm can be missed. It can replace a planned second look when clinical suspicion is low, but a negative scan does not cancel a second look when suspicion is high. Recidivism is either **residual** (incompletely removed) or **recurrent** (a new retraction), and both demand long-term follow-up.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 On call

CHRONIC EAR WITH A NEW SYMPTOM

1. New vertigo, facial weakness, headache, fever or neck stiffness: treat as a complication until proven otherwise.
2. Examine the facial nerve and do a fistula test.
3. Request CT, and MRI if intracranial spread is possible.
4. Start intravenous antibiotics and inform your senior.

8.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition** and structure: matrix, perimatrix, keratin.
- **Congenital, primary and secondary acquired,** and the theories of genesis.
- **Signs of the unsafe ear.**
- **Canal wall up versus down,** and who is suitable for each.
- **Fistula:** test, incidence and management.
- **Non-EPI diffusion MRI** for residual disease.

8.3 Common pitfalls

COMMON PITFALLS

- Treating an attic crust with drops and never lifting it.
- Forgetting the fistula test before surgery.
- Choosing canal wall up for a patient who cannot attend follow-up.
- Stripping matrix off a large fistula in an only hearing ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 19, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 11 and K. J. Lee ch. 22, 25. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Cholesteatoma of the Middle Ear

IMAGING FINDINGS

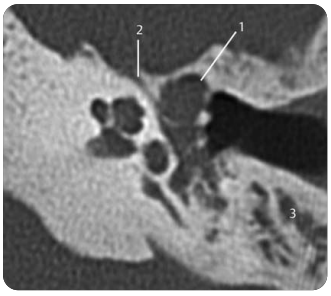
- Soft tissue in Prussak space (attic) or the posterior mesotympanum
- Blunted scutum; eroded, medially displaced ossicles
- Bright on non-EPI diffusion MRI; no enhancement

TOP DIFFERENTIALS

- Granulation tissue or effusion (no bone erosion)
- Cholesterol granuloma (bright on T1)
- Glomus tympanicum (enhancing, on the promontory)

CHECKLIST

- Five questions: tegmen, lateral canal fistula, facial canal, sinus tympani, ossicles
- Non-EPI DWI for residual or recurrent disease



CT, axial. The most frequently encountered mass in the middle ear is cholesteatoma, and therefore it is (also) shown in this chapter. In this…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · CONGENITAL

Congenital Cholesteatoma

IMAGING FINDINGS

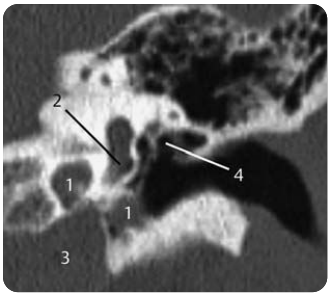
- Round, well-defined pearl behind an intact drum, usually anterosuperior mesotympanum
- Ossicles displaced or eroded, especially the incus long process and stapes
- Bright on non-EPI DWI; no enhancement

TOP DIFFERENTIALS

- Acquired cholesteatoma
- Glomus tympanicum
- Facial nerve schwannoma of the tympanic segment

CHECKLIST

- Intact drum and no history of otitis
- Look for mastoid and petrous apex extension



CT, coronal. CT excluded deeper invagination, but revealed other pathology. Coincidentally, an expansive infralabyrinthine…



Coronal temporal bone CT of the right ear demonstrates a large congenital cholesteatoma filling the middle ear cavity with subtle long process of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1026.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Labyrinthine Fistula due to Cholesteatoma

IMAGING FINDINGS

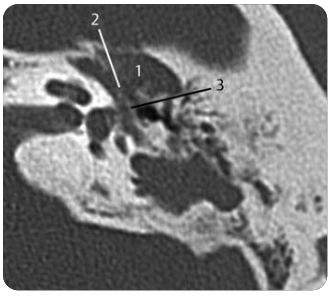
- Loss of the bony cover of the lateral semicircular canal next to cholesteatoma
- Best seen on coronal and Pöschl or Stenvers reformats
- Labyrinthine fluid on T2 may be reduced if labyrinthitis follows

TOP DIFFERENTIALS

- Partial volume of thin normal bone
- Superior canal dehiscence
- Post-surgical defect

CHECKLIST

- Report fistula site and size before surgery
- Confirm on thin slices in two planes



CT, axial. Note the smooth-bordered opacification suggestive of cholesteatoma in the anterior part of the middle ear (1), at the level of the…



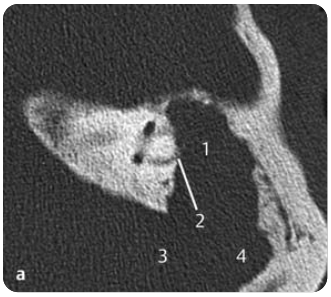
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Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Cholesteatoma with Intracranial Involvement

- Epidermoid cysts and cholesterol granulomas are more frequently located in the petrous apex.
- Endolymphatic sac tumor originates in the region between the posterior semicircular canal and the middle cranial fossa and sigmoid sinus, with typical imaging patterns (see separate sections).
- A meningo(encephalo)cele or meningioma demonstrates osseous destruction in the middle or posterior skull base, as well as continuity with the dural or intracranial structures.

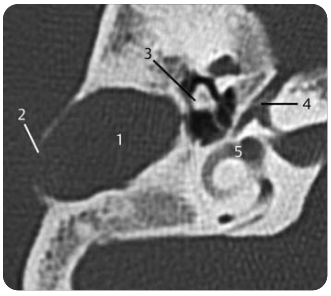


CT, axial. The middle ear was fully opacified with destruction of the ossicular chain. In this slice, the opacification of the epitympanic and…

EXTERNAL EAR & TYPANIC MEMBRANE · CONGENITAL

Inclusion Cholesteatoma and Atresia of the External Auditory Canal

- Expansion toward and/or destruction of the temporomandibular joint and toward the middle ear structures, abscess formation, and osteomyelitis and (intracranial) spread of infection.
- In case of aural atresia (first branchial groove anomaly), other dysmorphic features may be present too, especially in case of syndromal comorbidity.
- Particular attention needs to be paid to: the appearance of the middle ear cavity and mastoid pneumatization; signs of ankylosis or malformation of the ossicular chain; presence of inner ear deformities, the round and oval windows and the vestibular aqueduct; aberrations in the anterior and/or lateral course of the facial nerve, which may complicate surgery.



CT, axial. Expanding, round, smooth-bordered lesion (1) in the cranial part of the mastoid with partial destruction of the cortex (2). The head of the…